

Project ID :

25-26J-278

1. Topic (12 words max)

AI Mobile App for Early Detection of Student Mental Health via Behavior & Speech

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Mental health conditions like anxiety, depression, and stress-related terms are rapidly rising among students in higher education. Students in this cohort are often faced with significant academic challenges, navigate social adjustment, and personal transitions - all events that can affect the health of their mental well-being. With all the growing numbers it appears that a challenge persists in detection and where or how students might access the support required in times of distress. Students frequently do not seek help for a multitude of reasons including stigma, unaware of concerns, and/or unable to physically access a mental health professional deemed acceptable.

Additionally, several traditional means of detection, such as self-report questionnaire or traditional clinical interview, are subjective, sporadic, and reactive. There is a critical need for a well-scaled, proactive option that will assist in detecting potential early indicators of mental distress alert in real-time.

With the advent of AI and smartphones, we saw an opportunity for passive and continuous monitoring of behavioral and speech data that can help detect potential mental health risks. The goal of this study will be to create an AI-based mobile application that detects early signs of mental health disorders among university students utilizing multiple digital biomarker signals, including speech tone and patterns, screen time, typing behavior (especially as it relates to text messages), and sleep behavior.

Speech analysis is a very robust means to detect emotional states. During periods of depression or anxiety, students will exhibit certain features of speech such as, flat tone, slower speaking rate, irregular rhythm, longer pauses and not varying pitch [1]. The mobile application will capture data in real time by audio processing to identify these speech features either during phone calls or recorded through diary style memorandums to help see how students are faring emotionally [2].

Screen time is yet another important indicator. Too much or too little overall or selective screen time may represent behavior pulling (retreat) or behavior grinding (compulsive avoiding) behavior, both of which are prevalent in many mental disorders. A student's habits of how often they are and when they are interacting with their device might express a sense of routine or relatedness to psychological distress.

Typing behaviors (for text messaging) can provide relevant information to current and previous mental states. For example, students who are stressed/depressed might type slower (slower processing speed), use negative or vague emotional language and must edit more. These aspects of content and metadata (i.e., typing speed) might represent a shift in cognitive or emotional states, as well as potentially invoking previously seen patterns through the student's timeline data.

Sleep patterns, determined through device events (last event accessed prior to sleep and first activity after waking), are also important. More concerning sleep conditions linked with anxiety and depressive disorders are discontinuous sleep and inconsistent times for sleep

This application will serve as an early detection application and not a diagnostic tool. Notifying users when there is concern and providing self-care content and suggestions to see a mental health professional is important.

References (IEEE Format):

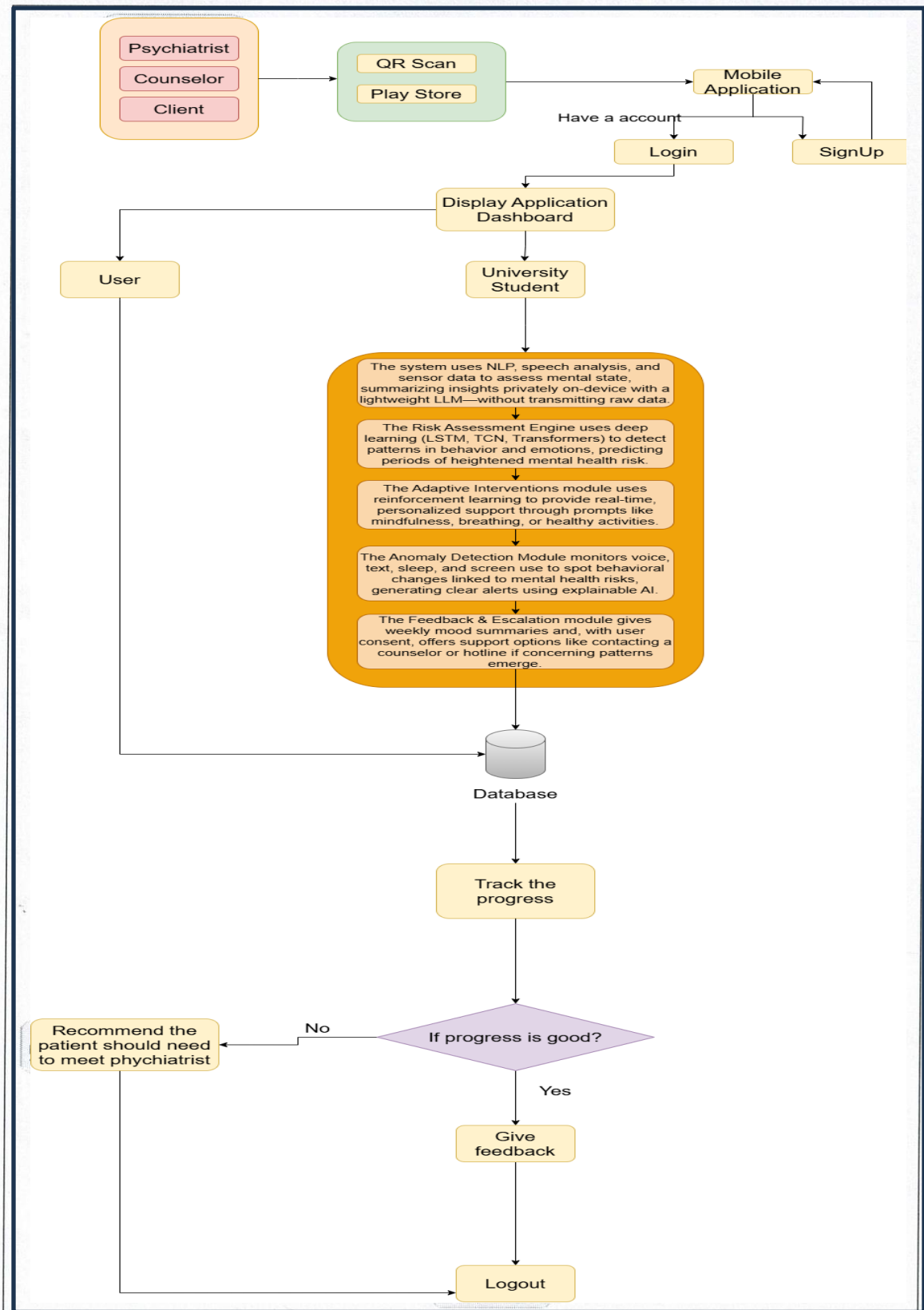
- [1] M. Saeb et al., "Mobile Phone Sensor Correlates of Depressive Symptom Severity in Daily-Life Behavior: An Exploratory Study," *J Med Internet Res*, vol. 17, no. 7, p. e175, 2015.
- [2] N. A. Cummins et al., "A Review of Depression and Suicide Risk Assessment Using Speech Analysis," *Speech Communication*, vol. 71, pp. 10–49, 2015.
- [3] D. C. Mohr, S. Zhang, and T. Schueller, "Personal Sensing: Understanding Mental Health Using Ubiquitous Sensors and Machine Learning," *Annu. Rev. Clin. Psychol.*, vol. 13, pp. 23–47, 2017.
- [4] T. Ghosh, A. Sharma, and S. Saha, "Analyzing Typing Patterns and Text Features for Mental Health Detection," *2021 IEEE Conference on Artificial Intelligence*, pp. 58–65, 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The solution is a mobile application that has a privacy-first strategy and leverages artificial intelligence for recognizing as well as reducing early signs of depression, anxiety, and stress in university students. The solution entails various modules in the system that offer real-time support and prediction. It features at its core an on-device AI engine that offers users' speech, behavior, and emotional tone summaries without sending the data to the cloud to ensure privacy.

The app utilizes temporal deep learning models (i.e., LSTM/CNNs) to detect patterns that precede mental health deterioration, enabling early risk identification. It also features a reinforcement learning-based intervention system that personalizes therapeutic content (e.g., relaxation prompts or CBT-based chatbots) based on individual user reaction and context. Furthermore, a multi-sensor anomaly detector fuses phone usage, speech, sleep data, and more to identify anomalous behaviors, offering users brief, human-readable explanations.

All processing is streamlined for on-device inference using TensorFlow Lite and federated learning for updating the model without compromising user privacy. A conceptual diagram illustrates the modules of the app: Data Collection → Summarization → Prediction → Intervention → Explanation. This architecture facilitates proactive, ethical, and personalized mental health care for each user.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

It takes interdisciplinary knowledge of clinical psychology, behavioral science, and human-computer interaction to develop this AI-based mental health app for college students. Team members must be conversant with validated screening tools (PHQ 9, GAD 7) and the diagnostic criteria for mood disorders (e.g., DSM 5 definitions of anxiety and depression). Designing interventions that are both successful and morally sound requires an understanding of cognitive behavioral therapy principles as well as the elements that affect students' well-being, such as social engagement, sleep patterns, and academic stressors. Furthermore, knowledge of user experience design guarantees that passive monitoring tools and active check-ins are combined in a way that promotes continuous participation without leading to survey fatigue.

Technically speaking, the project calls for a high level of expertise in mobile software engineering and artificial intelligence. Practitioners should be proficient in speech analysis and natural language processing, using transformer-based models (e.g., DistilBERT, wav2vec 2.0) for sentiment and emotion detection and time series modeling methods (e.g., Temporal Convolutional Networks or attention-based LSTMs) for customized risk prediction. Adaptive intervention policies require reinforcement learning frameworks (like Ray RLlib), while explainable AI toolkits (like Captum and SHAP), in conjunction with graph-based anomaly detectors, convert noisy sensor inputs into clear, useful insights. For Android devices built with React Native, mastery of on-device model optimization—quantization, pruning, and TensorFlow Lite deployment—is equally crucial to ensuring low battery impact and real-time responsiveness.

Lastly, the data requirements cover a variety of multimodal sources that represent the lives of students. Core training material includes speech corpora labeled for emotional state (e.g., DAIC WOZ, DEPAAC), behavioral logs recording location, screen time, and sleep (e.g., StudentLife), and student-specific surveys (e.g., attendance, assignment deadlines). Secure privacy protections are necessary for the ethical collection and handling of sensitive data. These include transparent consent workflows that comply with GDPR and HIPAA, secure on-device storage (encrypted SQLite), and federated learning frameworks (Flower or TensorFlow Federated) with differential privacy guarantees. Together, this technical know-how, data strategies, and domain skills will create a responsible, scalable, and successful student-focused mental health solution.

8. Objectives and Novelty

Main Objective: The main objective of this mobile application is to accurately assess each university student's current and future risk of depression—using a combination of passive behavioral and speech analysis alongside brief self-reports—and to deliver personalized, context-aware interventions that effectively reduce depressive symptoms before they worsen.

Member Name with Registration No	Sub Objective	Tasks	Novelty
De Silva A.H.H.	On-device AI summarization of speech, text, and behavior	• Collect speech transcripts, journal text, and behavioral logs • Run a lightweight LLM locally to fuse and summarize inputs • Store only concise summaries on the device	This module embeds a compact language model directly on the user's phone to continuously fuse raw speech transcripts, typed journal entries, and behavioral logs (such as screen time and typing patterns) into concise daily summaries. Rather than relying on cloud processing, it dynamically generates context-aware prompts tailored to each student's unique patterns—capturing emotional tone, stress indicators, and activity shifts—while ensuring that no sensitive data ever leaves the device. This approach not only preserves privacy but also delivers real-time, personalized

			insights that fuel downstream risk forecasting and interventions.
Kumarasinghe K.D.K.Y	Early risk prediction for future mental health decline	• Feed on-device summaries into a temporal deep-learning model • Forecast personalized risk windows days or weeks ahead • Continuously update predictions with new data	Leveraging temporal deep-learning architectures (e.g., attention-augmented LSTMs or Temporal CNNs), this component analyzes the on-device summaries to forecast individualized “windows of vulnerability” days or weeks ahead. It continuously refines its predictions as new behavioral and emotional data arrive, requiring minimal initial training data for each student. By proactively alerting users before their mental health deteriorates—based on personalized thresholds rather than generic cutoffs—the system empowers timely self-care actions and clinician outreach, shifting the paradigm from reactive support to preventative care.
Ranathunga A.K.M.	Personalized real-time interventions using reinforcement learning	• Monitor user engagement and context • Use an RL agent to choose the timing and type of prompt • Deliver mindfulness, breathing, or activity suggestions	A reinforcement-learning agent sits at the heart of this module, observing each student’s unique engagement signals—such as response times, exercise completion, and self-reported mood—and learning which type and timing of

			support yield the best outcomes. Over repeated interactions, it optimizes a mix of mindfulness prompts, guided breathing exercises, or activity suggestions (e.g., campus yoga sessions, brief social check-ins) to suit individual preferences and schedules. This continuous feedback loop ensures interventions become progressively more relevant, effective, and minimally intrusive.
Indrapala W.V.H.	Multi-sensor anomaly detection with human-readable explanations	• Fuse voice tone, text sentiment, sleep, screen time, and typing behavior • Detect deviations via graph-based models • Generate clear, explainable alerts	By constructing a dynamic graph over diverse sensor input voice tone metrics, text sentiment scores, sleep patterns, screen usage, and typing behavior, this module identifies atypical deviations that may signal emerging distress. When an anomaly is detected, it invokes explainable-AI techniques to translate the technical flag into clear, actionable language (e.g., "Your sleep duration dropped by 30% while vocal monotony increased for three days"). These transparent alerts help students understand exactly what changed and why it matters, fostering trust and guiding both self-help and professional intervention.



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Topic Assessment Form

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22032560	On-Device LLM Summarization. This component utilizes a compact transformer model optimized to run directly on students' smartphones, distilling free-form diary entries and speech transcripts into concise mood summaries in near real-time. It demonstrates expertise in Edge Computing and Mobile by applying model quantization and memory-efficient inference techniques, while seamlessly integrating with device sensors and microphones. From an AI/ML Engineering perspective, it involves knowledge distillation and transformer optimization methods to maintain accuracy under strict latency and resource constraints. The implementation requires careful software engineering to incorporate on-device orchestration and error handling within a mobile application framework. Overall, this novelty deepens skills in distributed AI architecture and mobile-first development.
IT22888334	Personalized Real-Time Interventions Using Reinforcement Learning. This component continuously monitors student behavior and dynamically adjusts coping strategies—such as mindfulness prompts or brief exercises—by training an RL agent that maximizes engagement and well-being over time. It exercises AI/ML (Reinforcement Learning) skills through the design of reward functions, policy learning algorithms (e.g., deep Q-learning or actor-critic), and simulation of student responses. From a Data Engineering & Distributed Systems standpoint, it requires building low-latency pipelines to ingest interaction metrics, manage agent state, and deploy updated policies across mobile clients. The Software Engineering & DevOps aspects include scalable backend microservices, CI/CD for model updates, and robust error handling to ensure uninterrupted delivery. Integrating Human-Computer Interaction principles, the system incorporates real-user feedback loops and A/B testing to iteratively refine intervention timing and content.
IT22321268	Early Risk Prediction for Future Mental Health Decline. Leveraging historical behavioral and speech data, this module employs time-series forecasting and classification models to predict students at risk of deteriorating mental health several weeks in advance. It showcases AI/ML (Predictive Analytics & Time-Series Modeling) expertise by implementing algorithms like LSTM networks or gradient-boosted decision trees tuned with cross-validation and hyperparameter search. On the Data Engineering side, it demands robust ETL pipelines for aggregating multi-modal features, handling missing data, and maintaining versioned feature stores. The Cloud Computing & DevOps specialization comes into play when deploying these predictive services as

	scalable APIs, monitoring model drift, and triggering retraining workflows. Finally, Software Engineering & Ethics considerations ensure transparent risk scores, explainable outputs, and compliance with privacy regulations when acting on sensitive predictions.
IT22917898	Multi-Sensor Anomaly Detection with Human-Readable Explanations. This feature fuses inputs like app usage patterns, mobility data, and vocal prosody to detect deviations from a student's baseline that may signal acute distress. It demonstrates Data Engineering proficiency by synchronizing heterogeneous time-series streams, performing feature extraction, and managing real-time anomaly-detection pipelines. In the realm of AI/ML (Anomaly Detection & Explainable AI), it applies algorithms such as autoencoders or isolation forests and integrates interpretability frameworks (e.g., SHAP or LIME) to generate concise, clinician-friendly rationales. The Visualization & HCI specialization is embodied in an interactive dashboard that presents anomalies alongside plain-language explanations, enabling mental-health professionals to quickly assess context. Underpinning all of this, Software Engineering & Cloud Infrastructure ensures reliable streaming, low-latency alerts, and secure data storage across platforms.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Loksha	Weerasinghe	
Co-Supervisor	Ms.	Ishara	Weerathunga	 27/06/2025
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	☒	No	☐
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- b) Does the proposed topic exhibit novelty?

Yes	☒	No	☐
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes	☒	No	☐
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	☒	No	☐
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

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Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.

